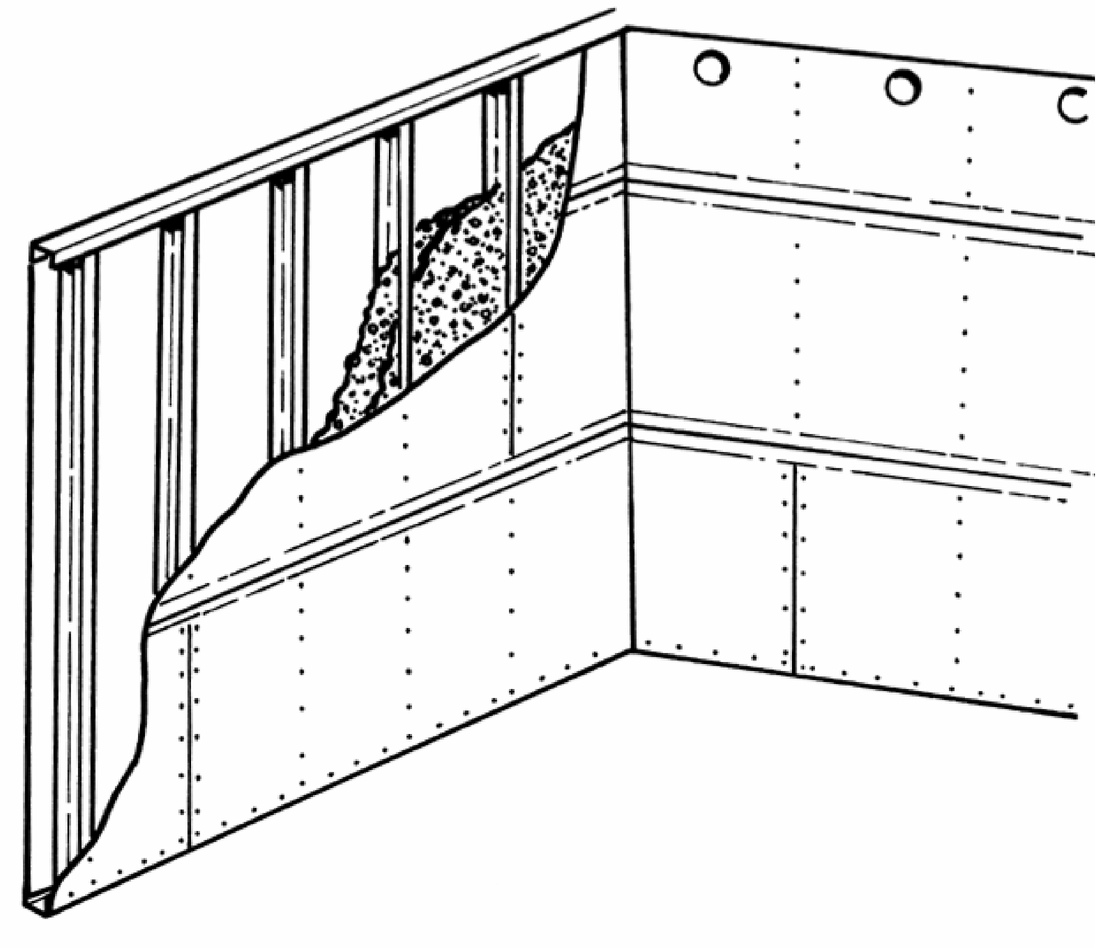
James Hardie

HardiWall™ Solid Wall System

For Internal Walls

SYSTEM SPECIFICATION



VERSION 2
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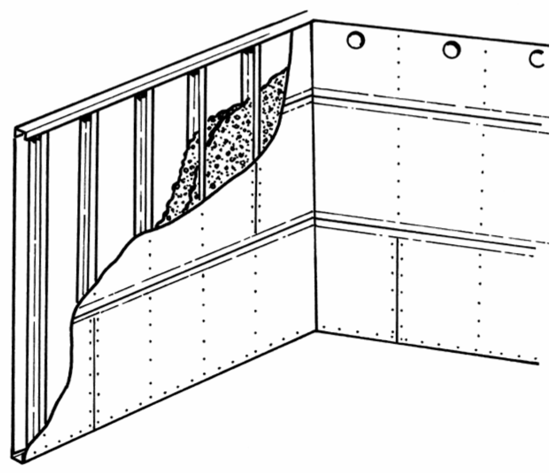
1. DESCRIPTION OF SYSTEM

The **Hardi**Wall™ Solid Wall System comprises light gauge galvanised steel stud framing, lined with **Hardi**Wall™ fibre-cement sheets and filled with a proprietary lightweight concrete core mix.



The **Hardi**Wall™ System is jointed using approved paper or fibre-glass gauze tape and the **Hardi**Stop™ Base Coat jointing compound, ready for skim plastering and painting.

Wall thickness is ultimately determined by the specifier to achieve aesthetic or fire rating elements. Varying the depth of the steel stud frame can change the wall thickness. The most common stud sizes provide finished nominal wall thickness of 77, 105 and 163mm. The finish wall tolerance is + or – 3mm across a length of 3 metres of wall.

2. SYSTEM COMPONENTS

2.1 Steel Framing

2.1.1 Stud
Section:
C-shaped lipped stud, 64mm deep and 35mm minimum wide for 77mm thick walls, 92mm deep and 35mm minimum wide for 105mm walls and 150mm deep and 35mm minimum wide for 163walls.

Service Holes:
Studs may have service holes pre-punched; otherwise holes should be created on site as required.

Gauge: minimum 0.8mm

Steel Grade: minimum 200MPa
Cold-rolled, hot-dip galvanised steel

2.1.2 Top and Bottom Track
Section:
Return flange U-shaped, having web depth compatible with those of studs of the same nominal size. Minimum width of flange shall be 32mm.

Gauge: minimum 0.8mm

Steel Grade: minimum 200MPa
Cold-rolled, hot-dip galvanised steel

2.2 Fibre-Cement Sheet

Brand:
HardiWall™ Sheet
(Manufactured by James Hardie)
Thickness: 6.6mm nominal.

Face finish: Sanded and surface sealed (gold trademark colour).

Edge treatment: Two long sides recessed.

SYSTEM SPECIFICATION

2.3 Sheet Fixing Screws

Brand:
HardiDrive™ Screws
(Supplied by James Hardie)

Description:

Self-embedding countersunk head winged screws, 25mm long, and 8mm nominal head size.

2.4 Core Fill Concrete

2.4.1 Core Mix Design

The core mix technology is a patented device and constitutes a number of base ingredients.

- Type GP cement or equivalent
- Sand - Dry
- Polystyrene Bead
- **Hardi**Add™ Core Mix Additive
- Clean potable water

2.4.2 Suggested Core Mix Delivery System

Mixer/hopper:	400-litre capacity planetary (drum) mixer connected to a hopper.
Pump:	Peristaltic (squeeze) type.
Delivery line:	Vertical – steel pipe with 50mm inner diameter equipped with couplings and bends. Horizontal – rubber hose 50mm inner diameter. (40 bar pressure rating)

2.4.3 **Hardi**Add™ Core Mix Additive

Supplier:	James Hardie
Packaging:	8kg (24 litre) bag

2.4.4 **Hardi**Flow™ Meter

This special flow meter has been selected by James Hardie to measure the workability and pumpability of the core mix. A reading between 6.0 and 7.0 units is considered a good workable mix.

3. HardiWall™ System Jointing

Base coat:	**Hardi**Stop™ Base Coat jointing compound (gold in colour) supplied by James Hardie.
Reinforcement:	Perforated paper jointing tape (recessed joints, butt joints and internal corners). Fibreglass self-adhesive gauze tape (optional). Galvanised steel or PVC corner angles (external corners).

4. HardiWall™ System Abutments

4.1 Wall-To-Wall / Wall-To-Ceiling

Abutment to Reinforced Concrete prior to wall plastering	Fill void between **Hardi**Wall™ Sheet and reinforced concrete using **Hardi**Stop™ Base Coat (by **Hardi**Wall™ installer); Render or plaster the RC surface up to the **Hardi**Wall™ surface and skim coat finish over the junction (by others);
Internal and external corners - **Hardi**Wall™ into **Hardi**Wall™	Internal Corners: **Hardi**Stop™ Base Coat + paper tape; and External Corners: **Hardi**Stop™ Base Coat + galvanised steel or PVC / plastic angles.

4.2 Wall-To-Column / Wall-To-Beam

Over Lap Joint (Butt Joint)	**Hardi**Wall™ setting-out and installation to provide finished wall datum. RC off-form surfaces should be made level by cut and plastering to produce level surface with **Hardi**Wall™ datum. The **Hardi**Wall™ Sheet is to overlap end stud by 10-15mm over RC surface. The RC surface is then plastered flush and a V-type notch made in the render alongside the edge of the **Hardi**Wall™ facing sheet to accommodate the **Hardi**Stop™ jointing system. This overlap joint is a standard joint option for the RC / **Hardi**Wall™ transition line. While the **Hardi**Stop™ jointing system is designed to resist cracking, the efficacy of the joint can be affected by excessive building movement.
Over-Sheeting Full face adhesive over-sheeting. Dot applied over-sheeting	**Hardi**Wall™ setting-out and installation to provide finished wall datum. RC off-form surfaces should be made level by cut and plastering to produce a gap tolerance of 2-4mm from plastered surface to rear of sheet. Using a 5mm notched trowel, apply approved full acrylic or polymer modified cement tile adhesive to plastered RC surface level surface with **Hardi**Wall™ datum. High build adhesive can be used directly onto off-form surface without the need for prior plastering to achieve 2-4mm tolerance. Apply **Hardi**Wall™ facing sheet to adhesive surface and make level with installed **Hardi**Wall™ surface and fix with sufficient concrete nails to hold the **Hardi**Wall™ facing sheet in place during the curing time for the adhesive. NOTE: Dot and full surface

	applied adhesive over-sheeting may result in hollow voids in over-sheeted areas. Tiles may be applied to over-sheeted areas providing sufficient masonry anchors are used. The number of fixings will be dependant on the nature of the tiles to be fixed.

5. HardiWall™ Wet Area Detailing

5.1 Bath Tub - General

Un-tiled areas under bath tubs:	It is recommended (but is included at the discretion of the specifier) that un-tiled areas from wall / floor junction to 100mm over the bath tub lip require the application of an approved water proofing agent to the **Hardi**Wall™ Sheet surface. This is to ensure that any water leakage from bathtub lip spills directly to the floor line and into an outlet in the event of sealant failure at the bathtub lip. The **Hardi**Wall™ Sheet surface itself is not affected by water but requires a waterproofing agent on un-tiled areas under the bath tub to ensure that the surface dries adequately with repeated / constant wetting. This will minimise potential for mould growth under the bathtub.
Tiled areas:	Application of water proofing agents to other areas is at the discretion of the specifier.

5.2 Bath Tub - Wall / Floor Junctions

Wall-to-floor Junctions:	Wall-to-floor junctions to be sealed during floor screeding and by application of cementitious fillet and approved flexible cementitious waterproofing systems by others.

5.3 Bath Tub - HardiWall™ / HardiWall™

Bath Tub Areas:	At **Hardi**Wall™ / **Hardi**Wall™ wall abutment, junctions where a 2-6mm gap is provided are to be sealed with **Hardi**Stop™ Base Coat (by **Hardi**Wall™ Installer). The application of **Hardi**Stop™ at the sheet gap junction does not constitute waterproofing. Waterproofing is achieved by the application of an approved waterproofing agent applied from the floor-to-wall junction to 1800mm above FFL at corners adjacent to the showerhead with 100mm min. legs extending from the corner centreline.

5.4 Bath Tub - HardiWall™ / RC

Junction Treatment:	At **Hardi**Wall™ / RC wall abutment, junctions where a 2-6mm gap is provided are to be sealed with **Hardi**Stop™ Base Coat (by **Hardi**Wall™ Installer) prior to plastering of the RC wall.

	The application of **Hardi**Stop™ at the sheet gap junction does not constitute waterproofing. Waterproofing is achieved by plastering the RC wall coating with approved waterproofing agent such that the waterproofing agent overlaps the HardiWall sheet surface. This plastering can be completed before or after un-tiled **Hardi**Wall™ surfaces are made good as per section 5.1. The overlap of waterproofing at **Hardi**Wall™ / RC junction must then be applied from the floor-to-wall junction to 1800mm above FFL at corners adjacent to the shower head with 100mm min. legs extending from the corner centreline.

6. INSTALLATION SEQUENCE

6.1 Stage 1 : – Site Readiness Prior to Installation by main contractor:

- The wall services (plumbing, electric conduits, etc.) shall be in position, but not fixed.
- Alignment of shear walls adjacent or connecting to the **Hardi**Wall™ System must be within the tolerance specified by the Installer of the **Hardi**Wall™ System - according to abutment detailing as set out in section 4.
- Floors shall be level and clean of debris, in a condition fit for laying out the bottom track of the wall. Where floors are inspected and found to be unsuitable for laying floor tracks, the main contractor must make good.
- Any other items, such as baths, must not be present in such a way as to obstruct the installation of the **Hardi**Wall™ System.

6.2 Stage 2 : – Framing

- Mark wall layout according to specified finished wall dimensions accounting for 2-3mm skim plaster and paint finishes.
- Anchor top and bottom tracks to ceiling and floor.
- Form door openings with double application of studs / C-Channel.
- Position studs between top and bottom tracks and leave floating.

6.3 Stage 3 : – Sheet Fixing (1st Side) – See below Stage 5

6.4 Stage 4 : – Installation of services.

- Fix services to the framing as required by others (plumbing, electric conduits etc.). – Work carried out by others.

6.5 Stage 5 : – Sheet Fixing – 2nd Side

- Measure and cut **Hardi**Wall™ sheets to size, providing cut-outs for openings for any existing services.
- Fix sheets with **Hardi**Drive™ screws onto the frame. Ensure both sides of the wall are fully sheeted. Masonry fixings and adhesive should be used to fix sheet onto concrete walls requiring over-sheeting. Sheets should be installed horizontally and staggered such that adjacent butt joints are not located on the same frame stud member.
- Provide appropriate fill holes between each pair of studs.

- Check screw fixings prior to pouring core infill concrete.

When fixing sheets, screws should be applied continuously from one side to the other or from the centre stud in an outwards direction to both sides.

Ensure that the self-embedding screw fixings DO NOT embed more than 1mm into the Hardiwall Sheet.

On fixing sheets, ensure screws embed through the main body of sheet, and not in the recess area of the sheet where ever possible.

6.6 Stage 6 : – Large Penetrations

Form penetration holes for sleeves provided by others. All works within sleeves by others.

6.7 Stage 7 : – **Core Filling**

- Prepare the proprietary core mix at the designated mixing station.

NB. The use of a ribbon blend mixer is recommended.

6.7.1 Workability Measure

It is necessary to measure mix workability to control the pour process. Measurement should occur at the beginning of each pour (first 2 batches) and thereafter at around 1 hour intervals. The measurement process is:

a) Wet the apparatus, ie **Hardi**Flow™ Meter, which comprises an outer partially perforated hollow tube and an inner calibrated measuring rod, and shake off all excess water.

b) Raise the measuring rod, and let it rest on the pin inside the apparatus.

c) Insert the apparatus slowly and vertically downward until the floater rests on the surface of the mix. Do not rotate the apparatus while inserting it into the mix.

d) After 60 seconds of complete insertion, lower the measuring rod slowly until it rests on the surface of the mix that entered into the hollow tube.

e) Read and record the level of the measuring rod to the nearest division.

f) A reading of 5.5 to 6.5 units represents an ideal mix for low-rise pumping (say up to 6 storeys). For higher-rise pumping the variability of the reading must be controlled between 6.0 and 7.0 units.

g) A reading of less than 5.5 units indicates that the mix is too dry. Water should be added in amounts of 500ml (approx. 2 cups) until an ideal result is achieved.

h) A reading of more than 7 units indicates that the mix is too wet. Cement should be added in amounts of 500g until an ideal result is achieved.

6.7.2 Workability Check (Visual)

A good workable mix should:

- Flow continuously down the discharge hopper and then self-level;
- Stick to the gloved palm of an overturned hand; and not display excess water sitting on the surface of a mixed batch.

6.7.3 Pumping

- Pump the core mix to the required floor through the delivery line and pour into the wall cavity.
- Inspect walls for voids and rectify (if necessary).

SYSTEM SPECIFICATION

- Wait for the core in-fill to dry (7 days minimum) before applying any material (including jointing compounds, adhesives, paint, etc.) to the wall face.

- Spills and seepages shall preferably be cleaned within 3 hours to 6 hours.

NB. The concrete core mix maybe poured on more than one occasion where the height of the wall precludes pouring in one pour.

6.8 Stage 8: HardiWall™ System Jointing

- For recessed, butt joints and internal corners of **Hardi**Wall™ / **Hardi**Wall™ junctions that are to be paint finished apply a first coat of **Hardi**Stop™ Base Coat followed immediately by a layer of reinforcing tape.

- On external corners firstly apply a coat of **Hardi**Stop™ Base Coat followed immediately by a galvanised steel or PVC/plastic angle.

- After 24 hours, apply a second coat of HardiStop™ Base Coat to the same areas but to a greater width.

- For deep holes in excess of 3mm (e.g. holes cut for core filling at the top of each stud bay), fill with plaster based filler and let set for 24 hours before applying a finishing coat of HardiStop™ Base Coat to the same areas in accordance with the above.

6.9 Abutments / Wet Area Detailing

Normally performed by other trades. – (Section 5)

6.10 General Finishing

Normally performed by other trades.

- For dry areas the surface is normally finished with a topcoat or skim coat, followed by painting. Note that **Hardi**Wall™ Sheet is supplied with a paint sealant already applied to the surface and that the **Hardi**Stop™ Base Coat compound is compatible with all common house (non-industrial) paints.

- All tiled areas, regardless of the surface to be finished, should incorporate:

  a). A flexible tile adhesive – either an approved acrylic or polymer modified cementitious adhesive; and

  b). A compressible grout – to accommodate for the long-term effects of tile growth.

  c). Tile abutments at all corners regardless of the substrate should incorporate a colour matched silicone or acrylic mastic to accommodate the effects of long-term tile growth.

- This represents best industry practice for tiling regardless of the substrate to be tiled.

SYSTEM SPECIFICATION

7. SYSTEM CONFIGURATION

Depending on specific project conditions, different installation configurations are possible in order to effect site efficiencies. These combinations are depicted in the table below.

Wall Thickness (mm)	HardiWall™ Sheet Thickness (mm)	Stud Frame Depth (mm)	Framing Centres Max. (mm)	Fixing Centres (mm)	Core Mix Fresh Density (kg/m³)	Suggested Core Filling Maximum Lift Height (mm)
77mm	6.6	64	300	150	550	2000
	6.6	64	300	100 - lower sheets 150 - upper sheets	800	2000
105mm	6.6	92	300	150	550	2000
	6.6	92	300	100 - lower sheets 150 - upper sheets	800	1500
163mm	6.6	150	300	150	550	1200
	6.6	150	300	100 - lower sheets 150 - upper sheets	800	1200

- **Framing**

For wall height 3 metres or below, the anchor fixing spacings for top and bottom tracks shall not be more than 300mm. For walls in excess of 3 metres high, please seek advice through the local Hardiwall distributor.

For walls in excess of 3 metres high or where the core mix density is in excess of 800Kg/m³, the maximum framing centres may need to be reduced to 250mm, please seek advice through the local HardiWall distributor.

For walls in excess of 3 metres high and which are to be subject to high loads or lateral loads e.g. granite/marble finishes, the thickness of the top and bottom tracks may have to be increased from 0.8mm to 1.5mm, please seek advice through the local Hardiwall distributor.

SYSTEM SPECIFICATION

- Sheeting lay-out — Horizontal, and staggered such that adjacent butt joints are not located on the same frame stud member.
- Core in-fill — Lightweight mix pumped into wall cavity.
- Jointing — Recess joint – **Hardi**Stop™ Base Coat + gauze or paper tape; and

  Butt joint – **Hardi**Stop™ Base Coat + gauze or paper tape.

8. TECHNICAL DATA SUMMARY

8.1 77mm Nominal Thickness HardiWall™ System

PROPERTIES	RATING	TEST STANDARD	TEST AUTHORITY
Weight (Nominal)	55kg/m^2		
Fire Resistance (FRP)	1 Hour	BS 476 : Part 20 & Part 22 – 1987	CSIRO - Division of Building, Construction and Engineering
Sound Insulation (Weighted Sound Reduction Index R_W)	39 SRI	BS 2760 : Part 3 - 0995 (ISO 140-3 : 1995) and BS 5821 : Part 1 - 1984 (ISO 717/1 - 1982)	Royal Melbourne Institute of Technology
Thermal Insulation	0.221 W/(m.K)	BS 874 : Part 2 : Section 2.1 - 1986	Building Investigation and Testing Services (Redhill) Ltd
Water Resistance	No penetration or leakage of water during and after test	BS 4315 : Part 2 - 1970	MateriaLab, Hong Kong

SYSTEM SPECIFICATION

8.2 105mm Nominal Thickness HardiWall™ System

PROPERTIES	RATING	TEST STANDARD	TEST AUTHORITY
Weight (Nominal)	70kg/m^2		
Fire Resistance (FRP)	2 Hours	BS 476 : Part 20 & Part 22 - 1987	CSIRO - Division of Building, Construction and Engineering
Sound Insulation (Weighted Sound Reduction Index R_W)	41 SRI	BS 2760 : Part 3 - 0995 (ISO 140-3 : 1995) and BS 5821 : Part 1 - 1984 (ISO 717/1 - 1982)	Royal Melbourne Institute of Technology
Thermal Insulation	0.2 W/(m.K) (est)	BS 874 : Part 2 : Section 2.1 - 1986	estimated on the basis of 77mm thick wall

8.3 163mm Nominal Thickness HardiWall™ System

PROPERTIES	RATING	TEST STANDARD	TEST AUTHORITY
Weight (Nominal)	130kg/m^2		
Fire Resistance (FRP)	4 Hours	BS 476 : Part 20 & Part 22 - 1987	CSIRO - Division of Building, Construction and Engineering
Sound Insulation (Weighted Sound Reduction Index R_W)	45 SRI (estimated)	BS 2760 : Part 3 - 0995 (ISO 140-3 : 1995) and BS 5821 : Part 1 - 1984 (ISO 717/1 - 1982)	Royal Melbourne Institute of Technology
Thermal Insulation	0.2 W/(m.K) (est)	BS 874 : Part 2 : Section 2.1 - 1986	estimated on the basis of the 77mm thick wall

SYSTEM SPECIFICATION

8.4 Strength & Robustness Testing by MateriaLab* of Hong Kong (77mm Wall)

PROPERTY	OUTCOME	BS 5234 : Part 2 : 1992:
(a) Stiffness	Load 500N Maximum Deflection = 0.20mm Residual Deflection = 0.01mm	Annex A
(b) Small hard body impact – surface damage	Depth of Indentation ≤ 1.5mm	Annex B
(c) Small hard body impact – perforation	No damage or perforation to the panel surface	Annex D
(d) Large soft body impact – damage	Permanent Deformation ≤ 1mm	Annex C
(e) Large soft body impact – structural damage	No collapse or dislocation of the panel;	Annex E
(f) Door slam	Res. Displacement = 0.45mm (3 slams); Res. Displacement = 0.71mm (100 slams);	Annex F
(g) Crowd pressure	No collapse or damage to the panel;	Annex G
(h) Lightweight anchorage – pull out	Applied Load = 100N; Pull-up shim plate was not released and no damage to the partition;	Annex H
(i) Lightweight anchorage – pull down	Applied Load = 250N; Pull-up shim plate was not released and no damage to the partition;	Annex J
(j) Heavyweight anchorage – wall cupboard	Applied Load = 4.0kN; Pull-up shim plate was not released and no damage to the partition;	Annex L

8.5 Hong Kong Housing Authority Testing by MateriaLab of Hong Kong

PROPERTY	OUTCOME	TEST METHOD:
Water Tightness	No water observable rear of sample	Hong Kong Housing Authority Test Methods
Tile Removal	No damage to the wall on removal of the tiles	
Chasing	No dislocation	

- MateriaLab is an accredited testing facility to HOKLAS